

Lab 1

Programing Lab With Python

BCACC393

1. Write a Python program to print Hello, World!.
2. Write a Python program to accept the user's name and display a welcome message.
3. Write a Python program to accept two numbers and calculate their sum, subtraction, multiplication, and division.
4. Write a Python program to calculate the area of a circle by accepting the radius from the user.
5. Write a Python program to calculate the area of a rectangle by accepting length and breadth from the user.
6. Write a Python program to convert temperature from Celsius to Fahrenheit.
7. Write a Python program to accept a number and display its data type using the `type()` function.
8. Write a Python program to demonstrate type casting by converting a string into an integer and a floating-point number.
9. Write a Python program to accept three numbers and calculate their average.
10. Write a Python program to accept a student's name and marks in three subjects and calculate the total and average marks.
11. Write a Python program to create a list of five numbers and display all the elements of the list.
12. Write a Python program to create a list of five numbers and display the first and last elements using indexing.
13. Write a Python program to create a list of numbers and add a new element using the `append()` function.
14. Write a Python program to create a list of numbers and remove an element using the `remove()` function.
15. Write a Python program to create a tuple of five numbers and display its elements.
16. Write a Python program to create a string and display its first character, last character, and length.
17. Write a Python program to create a list of five numbers and remove the last element using the `pop()` function. Display the list before and after using `pop()`.
18. Write a Python program to create a list of five elements and use `pop()` to remove an element at a specified index.

Learning Outcomes of this Lab 1:

- Students will be able to **understand and apply fundamental Python concepts** such as variables, input/output, data types, type casting, strings, lists, tuples, and basic operators.
- Students will be able to **write and execute simple Python programs** using functions and list operations such as `type()`, `append()`, `remove()`, `pop()`, and `len()`.